

A Mixed Reality Guidance System for Blind and Visually Impaired People

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ABSTRACT

Persons affected by blindness or visual impairments are challenged by spatially understanding unfamiliar environments. To obtain such understanding, they have to sense their environment closely and carefully. Especially objects outside the sensing area of analog assistive devices, such as a white cane, are simply not perceived and can be the cause of collisions. This project proposes a mixed reality guidance system that aims at preventing such problems. We use object detection and the 3D sensing capabilities of a mixed reality head mounted device to inform users about their spatial surroundings.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Human-centered computing—Accessibility—Accessibility systems and tools

1 INTRODUCTION

According to the World Health Organization (WHO) globally 295 million people suffer from severe visual impairment and 43 million are completely blind [7]. Blind or strongly visually impaired people (BVIP) typically use a white cane, a guide dog or the help of a guiding person to perceive the environment [4]. The white cane can detect obstacles and provide basic information like the noise of an object upon collision to surmise its substance. However, a white cane or guide dog do not provide any detailed object information. Especially the area above the user's pelvis is not supported by these guiding methods, leading to a higher risk of collisions [6]. This work presents an assistive Mixed Reality (MR) guidance system to support environment perception of BVIP.

2 RELATED WORK

Previous work improved the white cane with time-of-flight ultrasound sensing and by adding haptic feedback using a vibration motor accessed via a micro-controller board [1, 4]. Hashino et al. [2] estimate distances based on the order of received ultrasonic signals. However, recent studies pointed out weaknesses of such ultrasound-driven assistance applications due to the weak sensing performance [1]. In contrast to ultrasound sensing Zhang et al. [8] apply a deep transformer model on camera images to identify a walkable path and transparent objects, such as glass doors. For capturing the environment, smart vision glasses and a portable GPU were used and the user is informed via audio feedback. They further propose an approach for minor and moderate visually impaired people by using a Microsoft HoloLens Head Mounted Device (HMD) to highlight transparent objects.

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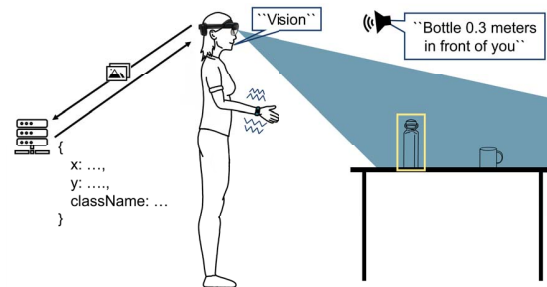


Figure 1: The user receives information about surrounding objects via haptic and audio feedback. Object detection is executed on a remote server, interexchanging information with the HMD.

3 APPROACH

In our work, we combine 3D spatial sensing of the HoloLens 2 with an object detection method to augment the spatial context for BVIP. The HMD captures and provides information about the environment and nearby objects in real-time. We process this data to provide information about the category and distance of the nearest object via audio feedback. Haptic feedback is included by using a smartwatch, an Apple Watch Series 6. Through vibrations varying on different distances, the user can sensibly perceive targeted objects. In contrast to [8] and other previous works, we do not intend to replace the white cane with our MR system. Instead our goal is to complement it, thus augmenting the user's spatial understanding capabilities. For example, when walking, the white cane is used for obstacle detection on the ground. The area of detecting objects using a white cane is limited. In unfamiliar environments the user has to detect everything above their pelvis using their hands. Through our MR system, no additional touches on various objects or surfaces are needed to gain understanding about the environment and users can safely navigate using our sensing system.

4 METHODOLOGY

The application is built using Unity3D and runs on a HoloLens 2 and an external server. Utilizing the hand tracking API of Microsoft's mixed reality toolkit, the user can interact with the system using a "Spiderman" type hand gesture, see Fig. 2. Besides that, the voice command "vision" can activate the MR guidance system. Furthermore, we use the spatial environment map captured by the HMD to provide haptic feedback on a wearable smartwatch. By varying haptic feedback, the user can get a spatial idea of the nearest object's position. The vibration intervals are grouped in three different strengths: low (distances $> 2m$), moderate (distances $1m - 2m$), high (distances $< 1m$). To obtain information on nearby objects the hand gesture or keyword activates the object detection. The current camera image is captured and sent to the server for object detection.

We use Yolo5s [3] to detect objects in the captured image with pretrained weights from the COCO dataset [5]. This dataset includes images from 80 object categories, including various everyday objects such as a *bottle* or a *cup*, and is thus well suited for the system.



Figure 2: Using the gesture or voice command the system is activated (left). Via the object detection algorithm the objects are identified (center). Using spatial mapping the nearest object to the hand of the user is identified (right).

To provide an access point for the HoloLens 2 application a Flask server is hosted providing a RESTful API. The 3D information from the spatial mapping of the HoloLens 2 and recognized objects from *Yolo5s* are combined. The x- and y-center coordinate of the detected bounding box are back-projected into the 3D world coordinate system. Utilizing the spatial map of the HoloLens 2 and applying a *sphere cast*, the 3D position of the object is detected. Based on the user's hand position, the nearest object is highlighted. To detect the nearest object, the distance and angle between the hand wrist and object are calculated. If the system detects a nearby object, an audio feedback provides the class name of the object as response and the distance in meters: *"There is a bottle 0.3 meters ahead from your hand."* If nothing is recognized in front of the user's hand, the feedback is: *"Nothing recognized two meters in front of you."*

5 PRELIMINARY RESULTS

We tested the initial prototype internally with healthy participants that had their eyes covered. In our first experiments, the subjects evaluated our approach detecting various objects like a *keyboard*, a *bottle* or a *cup* placed on a table, see Fig. 2. Our participants mainly judged the system as understandable and the haptic feedback was helping their orientation in the environment. The vibration could be more accurate. It was also noted that not varying between various vibration types (i.e., intervals) but varying the vibration intensity could help the haptic feedback. Additionally, information about the object shapes would be appreciated. Although the approach provides object category and distance, the distance in meters was recognized as not supportive enough. More day to day measurements like an "arm length" or "one step" were noted as easier to imagine.

We further measured the system's latency. The client-server communication consumes 400ms. 200ms are attributed to the server and object detection algorithm, while the remaining is attributed to the Wi-Fi and client. This latency was considered acceptable by our participants. Nevertheless, we plan to improve it.

6 DISCUSSION

In our first experiments, the combination of audio and haptic feedback was appreciated. Especially, the vibration to indicate the presence of objects proved to be positive. Vibration motors at various positions on the user's wrist could be used to extend this feedback. With varying positions of the haptic feedback additional information on the direct position of the object in front of the user's hand could be provided. We further aim to evaluate what gesture type could be most suitable for the activation and object selection, potentially executed by both left and right-handed people. Extending the command to the same hand that holds a white cane potentially frees the other hand. Another part of future work will be the extension of the object detection model to provide efficient 3D object detection. We further plan to conduct an evaluation with a BVIP target group. Finally, we want to follow up with studies on the social acceptabil-

ity of the approach. Higher social acceptance could be expected through the use of consumer hardware. For example, smartwatches are already being used in everyday life. However, if and how HMDs are accepted as assistive systems for BVIP in the public is yet to be investigated.

7 CONCLUSION

In this work, we present a novel MR guidance system for BVIP utilizing a MR HMD, a smartwatch and a state-of-the-art object detection model. Using our system, objects in unfamiliar environments can be detected safely. Additional information about the object category and distance to it is provided. Especially areas farther than the sensing area of a white cane or guiding dog can be explored. Future work will include the iterative improvement and a user study to verify that our system helps users in exploring unfamiliar environments in a safer manner.

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